

Reverse-Engineering AI Search

The complete GEO / AEO playbook — how platforms like HeyLead.ai, Profound, Otterly.ai and Peec.ai get businesses ranked #1 inside AI answers, and how to build a superior service yourself.

ChatGPT / SearchGPT

Perplexity

Google Gemini & AI Overviews

Claude

Copilot

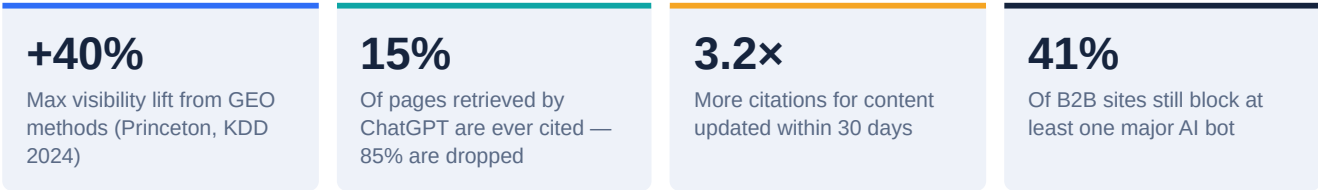
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Executive Summary

- **GEO (Generative Engine Optimization)** is the discipline of making a business the **answer** AI engines give — not just a link they list. The unit of competition has shifted from the ranked page to the **cited fact and the named brand**.
- **AI visibility is won off-site as much as on-site.** Brand mentions correlate with AI visibility at **0.664** — three times stronger than backlinks (**0.218**) — and earned media accounts for roughly **25% of all LLM citations**.
- **Each engine runs a different retrieval backbone:** ChatGPT/Copilot on Bing, Gemini on the Google index + Knowledge Graph, Claude on Brave Search, Perplexity on its own hybrid index. One-size-fits-all optimization is structurally blind.
- **Local queries are unusually winnable:** ~59% of local-intent prompts trigger live web search in ChatGPT (vs ~31% average), and Perplexity cites niche-authority domains with fewer than 10 referring domains 92.78% of the time.

- **The winning agencies are not "doing SEO better."** They engineer a cross-platform entity consensus — structured data, Wikidata, directories, reviews, digital PR — that LLMs can trust and quote.



01 Industry Landscape & Technical Foundations

1.1 GEO vs. Traditional SEO — Terminology & Mechanics

Table 1 — SEO vs. GEO — the operating model shift

Dimension	Traditional SEO	GEO / AEO
Target surface	Ranked list of blue links (SERP)	Synthesized answer with inline citations & brand mentions
Ranking unit	Page (URL)	Fact, claim, entity, extractable text chunk
Core signals	Keywords, backlinks, technical health	Entity clarity, factual density, third-party consensus, extractability
Success metric	Position, CTR, organic sessions	Share of AI Voice, citation rate, mention sentiment, AI-referral conversions
Index dependency	Google index	Bing (ChatGPT/Copilot), Google + KG (Gemini), Brave (Claude), Perplexity index
Keyword stuffing	Weak but historically worked	Performs worse than baseline in generative engines (Princeton GEO study)

Working definitions

- **GEO** — structuring content, entities and off-site signals so generative engines cite your pages and name your brand inside synthesized answers. Formalized by Aggarwal et al. (Princeton / Georgia Tech / IIT Delhi, KDD 2024): up to **+40% visibility lift**, led by Statistics Addition, Cite Sources and Quotation Addition.
- **AEO** — the overlapping discipline of being the direct answer (featured snippets → AI Overviews → chat answers). In practice the industry uses GEO/AEO interchangeably.
- **Share of AI Voice (SOV)** — the % of tracked category prompts where your brand is mentioned or cited vs. competitors. The GEO equivalent of rank tracking.

How RAG and vector search actually drive the answer

When a user asks *"Best health centers for seniors in Bern"*, the engine does not keyword-match. It runs a five-stage pipeline:

- 1

Intent parsing & query fan-out
The LLM decomposes the prompt into 2–4+ sub-queries (ChatGPT: 89.6% of prompts; Google AI Mode: 5–11). E.g. *"Alterszentrum Bern Bewertungen"*, *"nursing home Bern costs"*, *"Pflegeheim Bern Rangliste"*.

2 Retrieval

Sub-queries run against the engine's search index via BM25 (keyword) **and** dense vector retrieval (semantic embeddings — Perplexity uses its own pplx-embed models).

3 Reranking

Candidates scored on authority, freshness, entity relevance and structure. Perplexity runs a 3-layer ML reranker with a ~0.7 quality threshold — and discards the whole set and re-queries if too few results pass.

4 Citation selection

The model picks sentence-level facts and attaches the source that stated each fact most cleanly. Only 3–6 citations typically survive; **only ~15% of retrieved pages ever get cited**

5 Synthesis

The LLM writes the answer constrained by retrieved evidence, attaching inline citations.

KEY DISTINCTION — TWO VISIBILITY STATES

Mentions: brand named from training memory (no link; ChatGPT mentions brands ~3× more often than it cites them). **Citations:** clickable sources, only in browse/RAG mode. Track both separately.

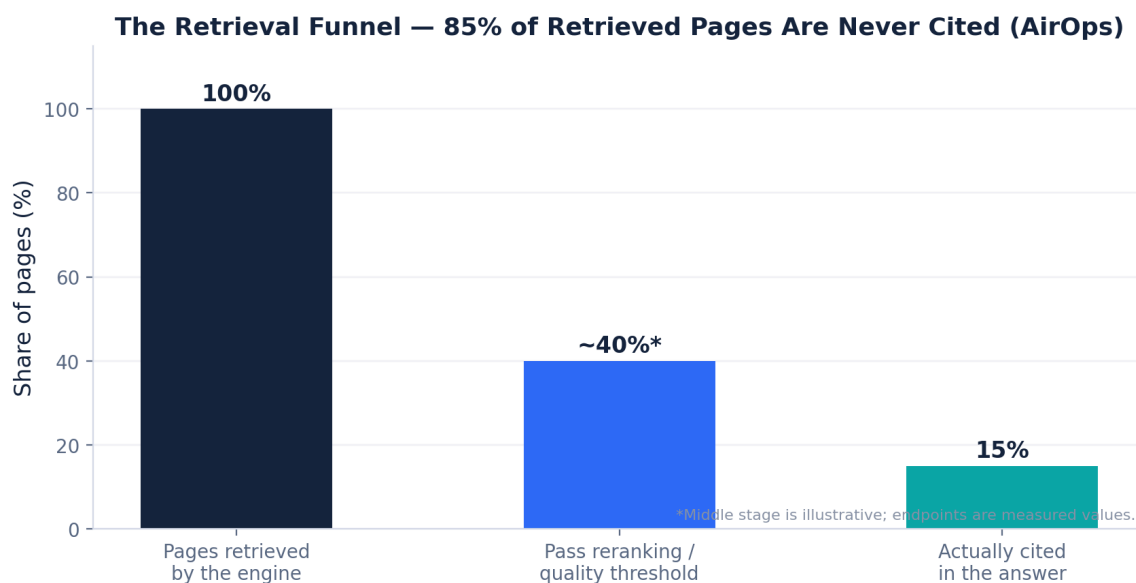


Figure 1 — The retrieval funnel — retrieval is not citation; 85% of retrieved pages are dropped

- **Vector search implication:** content must be the *semantic* best match for fan-out sub-queries you never explicitly targeted — a third of all cited pages enter via a fan-out query, not the original question. Topical cluster coverage beats single-page optimization.
- **Local advantage:** ~31% of ChatGPT prompts trigger web search overall — but **~59% for local queries**, making local GEO disproportionately winnable (Figure 2).

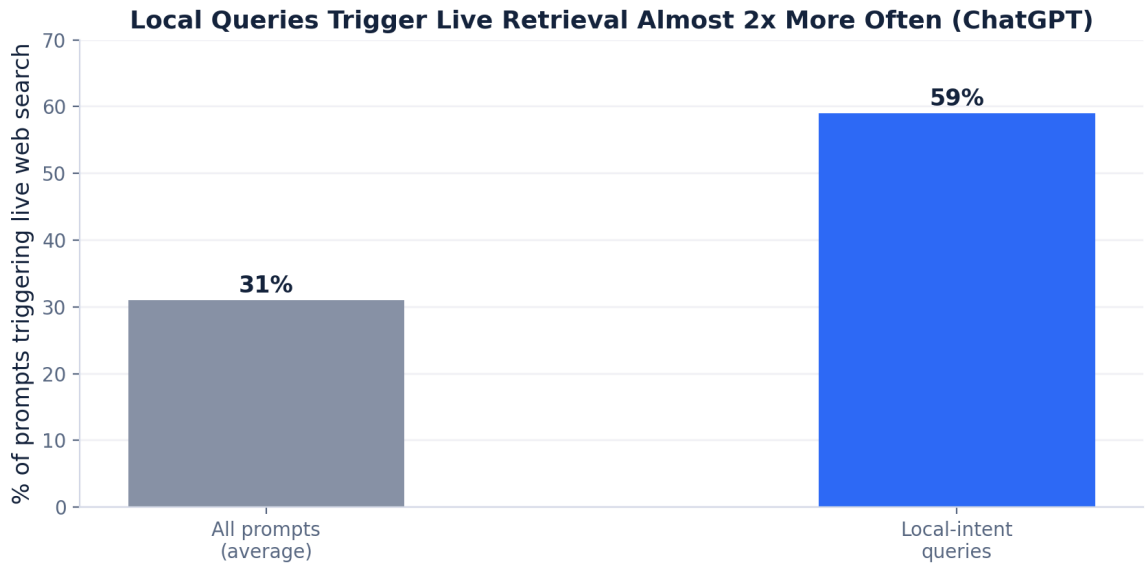


Figure 2 — Share of ChatGPT prompts that trigger live web retrieval, overall vs. local intent

1.2 The LLM Citation Ecosystem — Source Hierarchies per Engine

Each engine has a different retrieval backbone. A GEO strategy that ignores this is blind.

Table 2 — Master map — how each engine retrieves, grounds and cites

Engine	Retrieval backbone	Knowledge layer	Local query sources	Crawlers	Citation character
ChatGPT / SearchGPT	Bing index (primary) + OAI-SearchBot crawl	Training memory + live RAG	Bing Places, Yelp, Facebook reviews, TripAdvisor, directories	GPTBot · OAI-SearchBot · ChatGPT-User	Broad authority; Wikipedia & Reddit ~12–13% of citations each
Perplexity	Own index + partner APIs; hybrid BM25 + dense retrieval	Live RAG only — every answer grounded	Google Maps/GBP data, Yelp, Foursquare, TripAdvisor	PerplexityBot · Perplexity-User (ignores robots.txt)	Niche topical authority beats big domains; 92.78% of cited pages have <10 referring domains
Gemini / AI Overviews / AI Mode	Google index	Knowledge Graph (500B+ facts, 5B+ entities — Gemini is trained on it)	Google Business Profile, Maps, review text, GBP attributes	Googlebot (search + AIO) · Google-Extended (training token)	~92–94% of AIO citations from top-10 organic pages; AI Mode shares only 13.7% citation overlap with AIO
Claude	Brave Search index (encrypted live retrieval)	Training memory + web search tool	Practitioner blogs, deep /blog/ pages, listicles, documentation	ClaudeBot · Claude-SearchBot · Claude-User	Most selective: 56% of cited URLs are /blog/ paths, 47% listicles

Copilot	Bing index + Bing Places	Bing Knowledge + GPT reasoning	Bing Places, Facebook/Yelp aggregations	Bingbot; NOARCHIVE/NOCACHE meta as controls	Mirrors Bing ecosystem
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Cross-engine source tiers (what all engines disproportionately cite)

- 1 Reference / encyclopedic**
Wikipedia, Wikidata — structured, neutral, fact-dense.
- 2 Community / discussion**
Reddit, Quora — reads as first-hand experience. Note: volatile (Reddit swung ~60% → ~10% citation share within two weeks in late 2025).
- 3 Established media**
Editorial trust — Forbes, Reuters, NZZ; earned media ≈ 25% of all LLM citations.
- 4 Commerce / review / directory**
G2, Capterra, Yelp, Trustpilot, local directories. Sites with Trustpilot presence average 4.6–6.3 citations vs 1.8 without.

THE FOUR SOURCE TYPES BEHIND EVERY LOCAL AI ANSWER

(1) The business's own website · (2) Google Business Profile / Bing Places data · (3) Review platforms — read as *text*; AI quotes review language verbatim · (4) Directory & aggregator listings. **Conflicting facts across these sources (name, address, phone, hours, services) directly suppress recommendation confidence.**

What Actually Correlates With AI Visibility (Ahrefs, 75k brands)

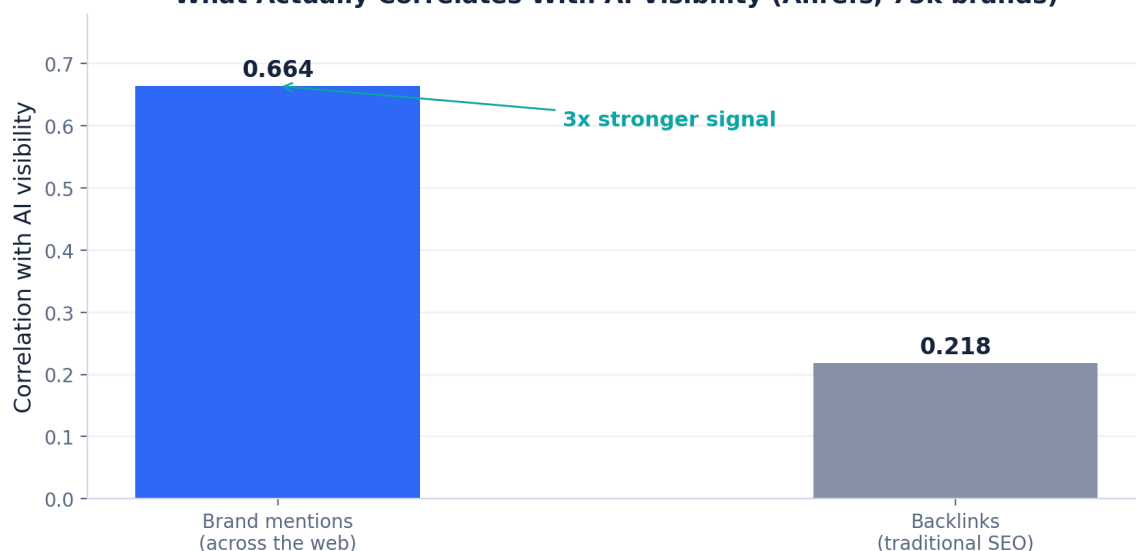


Figure 3 — Correlation with AI visibility: brand mentions (0.664) vs. backlinks (0.218) — Ahrefs, 75,000 brands

REVERSE-ENGINEERED WEIGHTING (DIRECTIONAL, NOT OFFICIAL)

Community consensus estimate: Authority & credibility ~40% · Content quality & utility ~35% · Platform trust ~25%. No vendor publishes its formula — treat all weightings in this report as directional and re-validate quarterly.

02 Reverse Engineering the Backend — What GEO Platforms Actually Do

2.0 The Platform Landscape

Table 3 — Who does what — the major GEO platforms and HeyLead.ai

Platform	HQ / backing	Model	What it actually does	Pricing signal
Profound	NYC · \$58.5M+ raised (Khosla, Kleiner, Sequoia)	Enterprise SaaS	Answer Engine Insights (visibility, SOV, sentiment, citations across 11 engines) · Prompt Volumes (real-user conversation panel) · Agent Analytics (AI bot traffic) · no-code Agents that draft/optimize content	\$99–499/mo entry · Enterprise \$2–5k+/mo
Otterly.ai	Persenbeug, Austria · bootstrapped	SMB SaaS	Prompt-library monitoring across 7 engines, 65+ countries/languages · citation & brand tracking, sentiment · crawlability checker, content audit, query fan-out tool · API/MCP/Looker Studio	From \$29/mo
Peec.ai	Berlin · \$29.1M Series A	Mid-market SaaS	Daily prompt execution per model/country · visibility score, position & sentiment · "used" vs "cited" source separation at domain/URL level · competitor benchmarking · Looker/API/MCP	From ~€89–95/mo
Scrunch	Salt Lake City · \$19M	SaaS + AXP	All 7 engines ungated · hallucination/misinformation detection · Agent Experience Platform	From ~\$250/mo
HeyLead.ai	Switzerland	Service + product hybrid	Free AI-visibility scan in a client portal (score per engine, gaps, strategy recommendations) → paid "AI Feed": 100–400 optimized pages published, backlinks built, old pages auto-refreshed on a compounding monthly cycle	Portal free · execution retainer

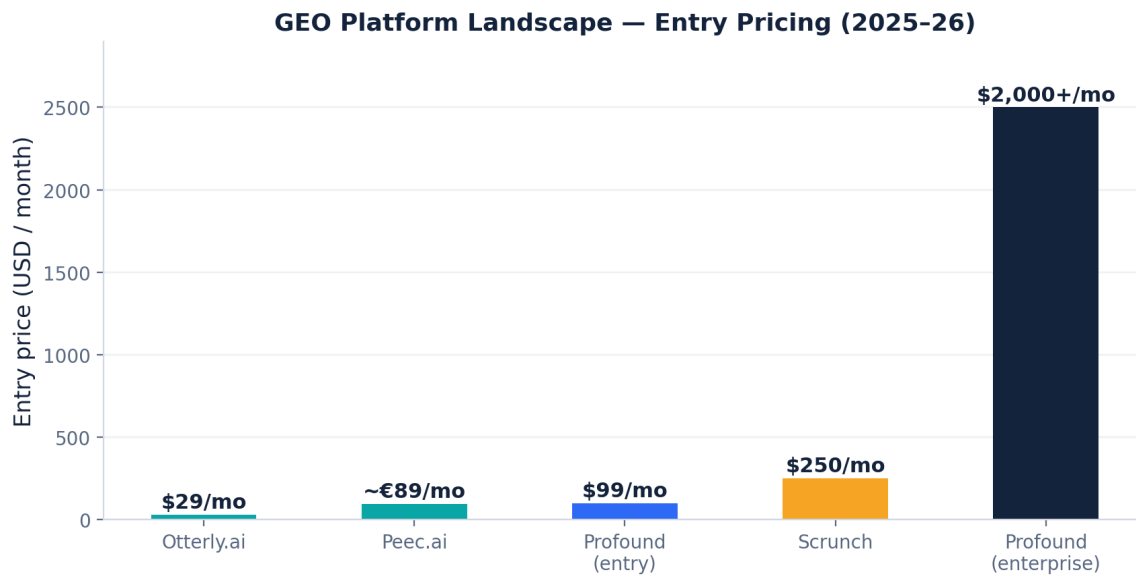


Figure 4 — Entry pricing across the GEO platform landscape

HEYLEAD.AI TEARDOWN — THE FUNNEL, MECHANICALLY

(1) Free measurement as lead magnet — automated multi-engine scan producing a score ("Overall 58/100, ChatGPT 72, Perplexity 41, 3 gaps"). **(2) The AI Feed** — programmatic generation of 100–400 structured pages (service × location × question permutations = fan-out coverage at scale). **(3) Compounding loop** — continuous deployment + backlink acquisition + automated refreshes, sold as "the longer it runs, the stronger it gets." Their published timeline: pages live months 1–2, first AI citations from month 3. **This is programmatic SEO rebuilt for RAG extraction, wrapped in a monitoring dashboard.**

2.1 AI Visibility Auditing & Benchmarking

The synthetic prompt testing method — the industry standard for measuring Share of Voice:

1 Prompt library construction

50–500 conversational prompts mirroring real queries, tagged by persona, funnel stage, language and region. Sources: keyword → prompt conversion, sales/support logs, People Also Ask, AlsoAsked, and (enterprise) Profound's real-conversation panel.

2 Scheduled multi-engine execution

Each prompt executed on every tracked engine on a fixed cadence (Peec: once/24h per model per country). Via official APIs (cheap, but answers differ from the consumer UI) or browser-fidelity capture (Profound's differentiator — the real rendered experience users see).

3 Response parsing

Per answer: brand mentioned? cited (URL linked)? position in answer (first recommendation vs footnote)? sentiment (enthusiastic / caveated / dismissed)? cited domains/URLs.

4 Metric computation

Brand coverage rate · Share of AI Voice (brand mentions ÷ total category mentions) · citation frequency per domain/URL · composite visibility score (HeyLead's "58/100") · LLM-classified sentiment score.

5 Gap analysis

Where competitors are cited and the client is not → which third-party domains to target, which content to build. Otterly ships this as "Recommendations"; Peec as source-level opportunity reports.

STATISTICAL HONESTY

LLM answers are non-deterministic — the same prompt yields different sources on different days. Serious platforms mitigate with daily repetition and trend-line reporting, never single snapshots. Evertune differentiates on rigor: base-model API access with 1M+ prompts/month per brand.

2.2 Knowledge Graph Alignment & Entity Optimization

The "anchor the business as a recognized entity" layer — the canonical four-layer stack plus connector:

Table 4 — The entity stack

Layer	Asset	Effort	AI weight
1	Crunchbase — legal entity anchor: founded, founders, category, HQ	1–2 h	Read by ChatGPT, Perplexity, Gemini, Claude
2	LinkedIn Company Page — corroboration layer	2–4 h	Read by all major engines
3	Wikidata — structured, queryable QID; no notability requirement , any legitimate business can enter	4–8 h	Heaviest on Gemini; moderate elsewhere; every major AI training pipeline ingests it
4	Wikipedia — only with genuine press coverage (notability-gated)	Variable	Heaviest weight across all engines
⊕	sameAs JSON-LD — wires all profiles to the website entity	1–2 h	Read by every crawler that parses JSON-LD; officially supported by Google

- **Entity home.** Designate one page (usually About) as the canonical entity anchor with `@id` and full Organization/LocalBusiness JSON-LD. Google's Knowledge Panel description is increasingly pulled from this page when no Wikipedia entry exists.
- **Wikidata entry.** Complete properties (inception, HQ, industry, official website, GLN/UID for Swiss companies) with references.
- **knowsAbout property** — severely underused; declares topical authority domains to the Knowledge Graph.
- **Entity consistency deployment** — standardized name, description, category and NAP across 50+ sources; entity ambiguity measurably reduces AI confidence.
- **Entity linking in content** — `mentions` schema pointing to Wikidata entities for discussed concepts.

THE PAYOFF CHAIN

Entity establishment → Knowledge Graph inclusion → Gemini training data → AI Overview / AI Mode citations. Because **Gemini is trained on the Knowledge Graph**, brands with resolved entities hold a structural advantage.

2.3 Citation & Digital PR Engineering

The off-site layer — inverting traditional link-building math:

95%

Of AI-cited links come from earned (non-paid) sources — Muck Rack, 1M+ citations

+239%

Median brand-citation lift from syndicating articles across third-party news sites (Stacker × Scrunch, 2026)

5×

Growth of wire-distributed press releases as AI citations (Jul → Dec 2025)

The six compounding PR tactics agencies run

- **Original research / proprietary data studies** — the single strongest citation magnet (statistics addition is also the #1 on-page GEO factor).
- **Expert commentary** via HARO-successors, Qwoted, Featured — with structured-format pitches.
- **Wire syndication** — PR Newswire is a top-20 AI-cited domain.
- **Executive thought leadership** with Author schema on bylines.
- **Reactive newsjacking.**
- **Trade bylines** in vertical publications — for DACH: Heise, t3n, NZZ, Handelszeitung (German-language AI answers lean on DACH-specific media).

Community & directory seeding

- **Reddit / Quora:** genuine, non-promotional participation — among the most-cited domains on ChatGPT and Perplexity, but volatile and spam-sensitive.
- **Review platforms:** Trustpilot presence alone moves averages from 1.8 → 4.6–6.3 citations. For local: Facebook and Yelp dominate Bing Places review sourcing (Facebook 716 vs Yelp 482 vs TripAdvisor 217 occurrences in Whitespark's 153-query study) — and **Bing Places feeds ChatGPT local recommendations.**
- **Niche directories:** category-specific listings (for Swiss healthcare: comparis.ch, local.ch, search.ch — Section 4).
- **Bing index optimization:** sitemap to Bing Webmaster Tools + claim Bing Places — the most-overlooked ChatGPT lever, since ChatGPT retrieval runs on Bing infrastructure.

2.4 Semantic Content Structuring — Writing for RAG Extraction

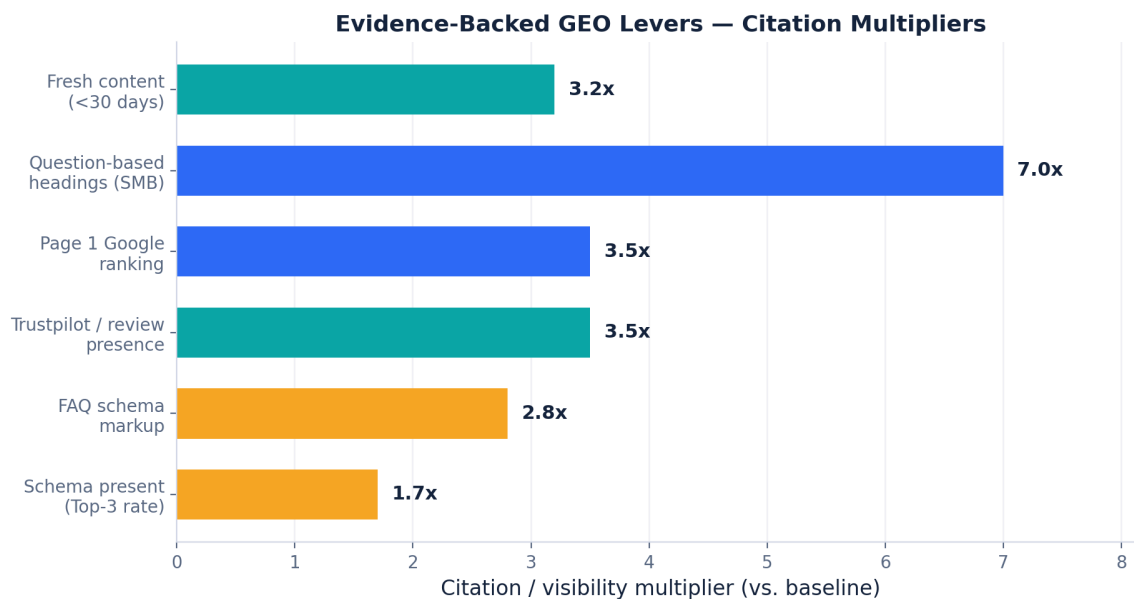


Figure 5 — Measured citation multipliers for the highest-impact content and schema levers

Structural rules (evidence-backed)

- **BLUF — answer in the first 60–100 words.** 90% of top Perplexity citations follow this; 44.2% of all LLM citations come from the first 30% of a page.
- **Chunk-level self-containment.** Each H2/H3 section understandable in isolation — AI extracts discrete answer chunks, not whole pages. Lead sections with a 1–2 sentence direct answer.
- **Question-based headings** mirroring fan-out sub-queries — ~7× citation impact for smaller domains.
- **Factual density:** specific statistics, named sources, dates, direct quotes — the Princeton top-3 methods.
- **Freshness:** content updated <30 days gets **3.2× more citations**; monthly refresh is a non-negotiable retainer task.
- **Tables & structured comparisons** — machine-liftable formats win citation selection over prose.
- **EAV clarity:** state facts as standalone declarative sentences ("The center offers 120 residential places, 24/7 nursing care, and specialized dementia units in Bern.") — the model lifts the sentence that needs no editing.
- **E-E-A-T surface signals** for YMYL (health/legal/finance): author credentials, methodology, awards — pulled from across the web, not just your site.

SCHEMA STACK (JSON-LD, VALIDATED)

Pages with schema show a **47% Top-3 citation rate vs 28%** without; FAQ-schema pages are **~2.8×** more likely cited. The local stack: LocalBusiness (most specific subtype) + Service + FAQPage + AggregateRating/Review + Organization + BreadcrumbList. Full template in Section 4.

- **Topic cluster architecture:** pillar + cluster pages covering every fan-out branch (costs, comparisons, alternatives, "vs", location variants) with contextual internal linking — this is how you enter the candidate pool through multiple sub-queries. **This is precisely what HeyLead's 100–400 page "AI Feed" industrializes.**

- **Technical rendering:** AI crawlers are poor JavaScript executors — core content must exist in server-rendered HTML.

2.5 Crawler Management — robots.txt & Indexation

41% of B2B sites still block at least one major AI bot — usually leftover 2023 "block everything" rules — costing an estimated 18–34% of potential citations per blocked engine.

Table 5 — The AI bot matrix — every provider runs two bot families

Bot	Operator	Function	Block cost
GPTBot	OpenAI	Training	No search visibility cost
OAI-SearchBot	OpenAI	ChatGPT search indexing	Critical You vanish from ChatGPT answers
ChatGPT-User	OpenAI	Live user-initiated fetch	May ignore robots.txt
ClaudeBot	Anthropic	Training	No direct citation cost
Claude-SearchBot	Anthropic	Search retrieval	High Reduced Claude visibility
PerplexityBot	Perplexity	Search retrieval (pure)	Critical No Perplexity citations
Perplexity-User	Perplexity	Live fetch	Openly ignores robots.txt
Google-Extended	Google	Control token, not a crawler — controls Gemini training only	None Zero effect on rankings or AI Overviews
Applebot-Extended	Apple	Control token (training opt-out)	None on search

Recommended default policy for GEO clients

Allow search/retrieval bots, decide training bots per IP strategy, keep sensitive paths blocked for all, always reference sitemaps:

```
# --- AI search/retrieval bots: ALLOW (citation eligibility) ---
User-agent: OAI-SearchBot
Allow: /
User-agent: ChatGPT-User
Allow: /
User-agent: PerplexityBot
Allow: /
User-agent: Claude-SearchBot
Allow: /

# --- Training bots: business decision ---
User-agent: GPTBot
Allow: /
User-agent: ClaudeBot
Allow: /
User-agent: Google-Extended
```

```
# --- Universal protections ---
User-agent: *
Disallow: /wp-admin/
Disallow: /checkout/
Disallow: /account/

Sitemap: https://yourdomain.ch/sitemap.xml
```

COMMON AUDIT FINDINGS

- Leftover `Disallow: /` under GPTBot
- Allowing GPTBot but forgetting OAI-SearchBot (kills ChatGPT citations anyway)
- **Cloudflare's "Block AI Bots" toggle overriding robots.txt silently**
- noindex on pages meant to be cited · sitemap path blocked

ENFORCEMENT REALITY

robots.txt is advisory. Perplexity was documented using undeclared rotating crawlers (Cloudflare, Aug 2025); ChatGPT-User and Perplexity-User openly disregard it. Hard stops require WAF/IP-range rules — all major vendors publish machine-readable IP lists. Verify real bots via reverse DNS.

LLMS.TXT — THE HONEST VERDICT

SE Ranking's 300k-domain study found **no statistically significant correlation with AI citations**; Otterly's 90-day experiment saw only 0.1% of AI bot traffic touch the file; no major provider has committed to it. Verdict: 30 minutes to implement, near-zero downside — **but it is not a ranking lever. Don't sell it as one.**

03 Step-by-Step Service & Product Blueprint

3.1 The "AI Search Readiness & Ranking Audit" — Your Front-End Product

Package it exactly like HeyLead's free portal scan: low-friction, high-perceived-value, converts prospects. Five modules:

Table 6 — The five audit modules

Module	What you do & deliver
A • Visibility Baseline score	Build a 30–60 prompt library (category, best-of, comparison, location queries × languages). Execute across ChatGPT (search on), Perplexity, Gemini, AI Overviews, Copilot in clean sessions with screenshots. Score per engine: coverage, position, sentiment, cited sources — plus the same runs for 3–5 competitors.
B • Citation Source Map	Extract every cited domain/URL; cluster into directories, review platforms, media, community, competitor sites. Output: the exact third-party properties the client must be present on → the off-site execution backlog.
C • Technical AI Readiness	robots.txt vs the full bot matrix · Cloudflare AI-bot toggle · noindex leaks · Bing Webmaster Tools status (sitemap? indexed?) · server-rendering check · server-log analysis for GPTBot / OAI-SearchBot / ClaudeBot / PerplexityBot visits · full JSON-LD inventory.

D • Entity & Consensus Knowledge Panel? Wikidata QID? Crunchbase claimed? GBP/Bing Places claimed? NAP consistency scan across all listings (conflicts = suppressed AI confidence) · review footprint: volume, velocity, rating, text content per platform.

E • Content Extractability BLUF check · chunk self-containment · question headings · factual density · FAQ coverage vs prompt library · freshness dates · query fan-out gap analysis (run priority prompts, record the sub-queries engines generate, map against existing content).

DELIVERABLE

A scored 100-point report across the five modules + prioritized gap list + 90-day proposal. **The audit is your sales weapon** — the score creates urgency, the gap list creates the retainer.

3.2 The 30-60-90 Execution Roadmap — Local Business Playbook

Table 7 — Days 1–30 — Foundation & Access

Week	Workstream	Exact tasks
Week 1	Technical unblocking	Fix robots.txt (allow OAI-SearchBot, PerplexityBot, Claude-SearchBot, ChatGPT-User) · check Cloudflare/CDN bot rules · Bing Webmaster Tools: verify, sitemap, IndexNow · claim Bing Places · fix server-side rendering · XML sitemaps
Week 2	Entity anchoring	Complete Google Business Profile (every attribute, 20+ photos, weekly posts) + Bing Places · claim Crunchbase, align LinkedIn · create Wikidata entry · entity-home page with full JSON-LD + sameAs + knowsAbout · NAP cleanup
Week 3	Schema & core pages	Deploy full schema stack (Section 4 template) · rebuild top 5 money pages: BLUF + chunk structure + question headings + stats/quotes + visible dates
Week 4	Directory & review foundation	Claim all category directories · launch permanent review acquisition: trigger at service completion, direct link, 2–5 new reviews/week, respond to 100% · seed review text with service keywords naturally

Table 8 — Days 31–60 — Content & Citation Engineering

Workstream	Exact tasks
Cluster deployment	Pillar + 15–40 cluster pages covering the fan-out map (costs, comparisons, alternatives, neighborhood variants, per language). HeyLead-style: 100+ structured pages — but every page needs genuinely unique value, not doorway spam (Google scaled-content abuse policy).
First PR wave	One proprietary data story (e.g. "Senior care cost index Bern 2026") → wire syndication + Swiss media outreach + trade byline.
Listicle & directory seeding	Get the client into every relevant "best of" list, comparison page and niche directory from the Citation Source Map.
Community presence	Genuine participation in relevant subreddits/local forums; expert Quora answers.

Refresh & monitor	Refresh all money pages (30-day freshness cycle begins) · stand up the monitoring dashboard (3.3) · weekly tracking begins.
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Table 9 — Days 61–90 — Compounding & Proof

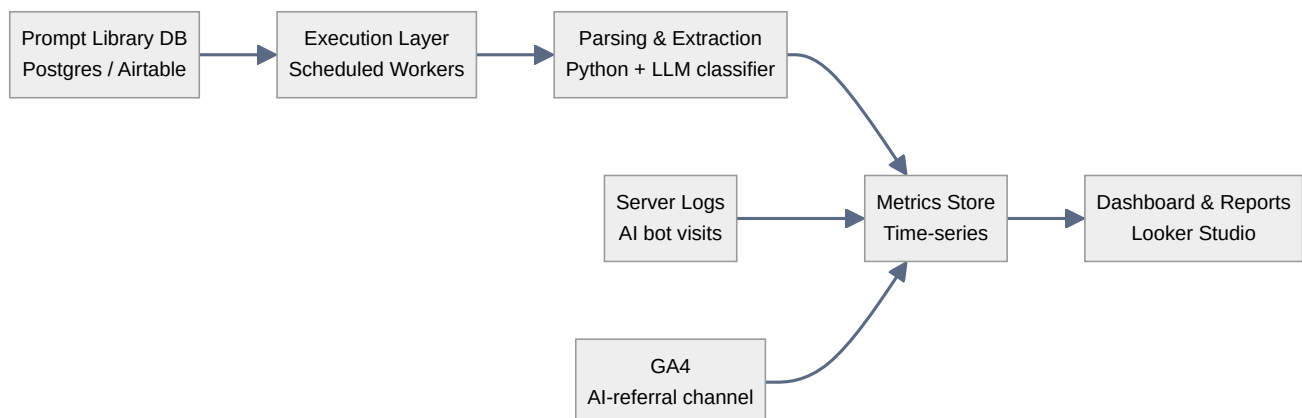
Workstream	Exact tasks
PR & content iteration	Second PR angle + wire release · two newsjacking moments · close remaining fan-out gaps discovered from monitoring data.
Review doubling-down	Focus on the platforms that appear in <i>your category's</i> AI answers (check weekly which sources engines actually cite).
Multilingual wave	DE pages live → FR localization with proper hreflang (Section 4).
ROI report	SOV trend per engine · citation count trend · AI-referral traffic (GA4 custom channel for chat.openai.com / perplexity.ai / gemini / copilot referrers) · lead attribution. Publish the case study — it sells the next ten clients.

ONGOING RETAINER RHYTHM

Weekly: monitoring & review responses → Bi-weekly: content refresh → Monthly: new cluster pages + reporting → Quarterly: entity/directory expansion + robots.txt review.

3.3 The Software Architecture Blueprint — Your Own Monitoring Stack

Replicate 70–80% of Otterly/Peec functionality in-house:



- **Execution layer:** scheduled workers (n8n / GitHub Actions / Cloud Run, daily or weekly cron) calling OpenAI API (web_search tool) → ChatGPT · Perplexity Sonar API → Perplexity · Gemini API (Search grounding) → Gemini. Optional headless-browser capture (Playwright) for consumer-UI fidelity. Run each prompt 1–3× per engine per cycle.
- **Parsing:** brand mention (regex + fuzzy match + aliases) · cited URLs/domains normalized · mention position · competitor mentions · sentiment via a small LLM classifier call.
- **Dashboard & reporting:** Looker Studio (free, client-facing) or Streamlit/Retool internal · automated weekly white-label reports · alerts on mention lost, sentiment drop, new competitor.

KEY ENGINEERING DECISIONS

- **API vs browser capture:** APIs cheap/stable but differ from consumer UI; browser capture is ground truth but fragile. Best: API daily + browser spot-checks weekly.
- **Non-determinism:** never report single runs; rolling averages over ≥ 7 daily runs.
- **Cost:** 50 prompts $\times$ 4 engines $\times$ daily $\approx$ 6,000 calls/month — trivially cheap. The value is parsing + presentation.

POSITIONING VS THE \$29–95/MO TOOLS

Don't compete as a tool — **bundle monitoring inside your execution retainer as proof-of-work.** The dashboard is your retention mechanism, not your product. ROI KPIs: SOV per engine · mention volume & sentiment · evaluation-stage inclusion ("best/vs/alternatives") · AI-referral sessions · AI-attributed leads · cited-domain coverage growth.

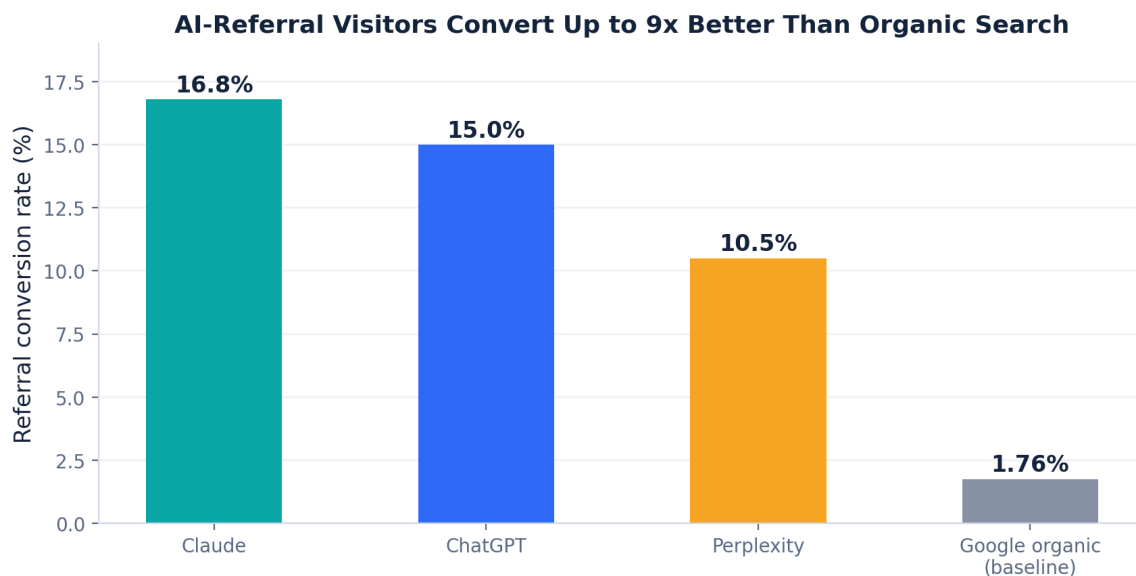


Figure 6 — The ROI story — AI-referral visitors convert up to 9x better than Google organic

04 Case Study — "Best Health Center for Elders in Bern"

Target query family (trilingual): **DE** "*Bestes Altersheim / Pflegeheim in Bern*" · **FR** "*Meilleur EMS / centre de santé pour seniors à Berne*" · **EN** "*Best health center for elders in Bern*". Hypothetical client: **"Sonnenhof Bern"** — private senior residence with nursing, dementia and day-care services.

4.1 How Each Engine Will Build Its Answer

Table 10 — Per-engine retrieval for the Bern eldercare query — and where Sonnenhof must exist

Engine	What it retrieves for this query	Where Sonnenhof must exist
ChatGPT	Bing results + fan-outs (" <i>Pflegeheim Bern Bewertungen</i> ", " <i>Altersheim Bern Kosten</i> ") + Bing Places + review aggregations (Facebook, Yelp)	Bing index · Bing Places · Facebook reviews · directories · DE/FR/EN pages

Gemini / AIO	Google index + Knowledge Graph + GBP data + review text + Maps	Complete GBP · Google reviews · KG entity · top-10 organic pages
Perplexity	Own index + live crawl of top 10–20 results; local data from Maps/Yelp-type sources; ~5–10 sources cited	Crawlable BLUF pages · comparis/local.ch profiles · fresh content
Claude	Brave Search; prefers deep blog/guide pages, listicles, dated content with citations	Guide-style content ("Choosing a senior residence in Bern: costs, care levels, questions to ask")
Copilot	Bing + Bing Places	Same as ChatGPT

MULTILINGUAL MECHANICS

German AI answers lean on DACH sources; French answers on FR-language sources. DE and FR page versions are treated as **separate sites** by Google — proper `hreflang`, localized (not machine-translated) copy, and per-language directory presence are required. A .ch domain, Swiss hosting and .ch backlinks strengthen the geo-signal for all engines.

4.2 End-to-End Strategy (Mapped to the 30-60-90)

Phase 1 — Be findable & unambiguous (Days 1–30)

1 Technical access

robots.txt + Bing Webmaster Tools + Bing Places (CH categories: "Pflegeheim", "Alters- und Pflegezentrum").

2 GBP mastery

Primary category *Nursing home / Altersheim*, every attribute (wheelchair access, dementia care), 20+ photos, weekly German posts.

3 Entity stack

Wikidata entry (instance of: nursing home; location: Bern; official website) + entity-home page with the JSON-LD below.

4 Consensus layer

NAP cleanup + Swiss directory claims (4.3) + rebuild the 5 core pages per language with BLUF structure — e.g. the dementia page opens: *"Sonnenhof Bern operates a specialized 28-bed dementia unit with 24/7 nursing staff, secured gardens and a 1:4 caregiver ratio in Bern-Bümpliz."* — a liftable EAV sentence.

```
{
  "@context": "https://schema.org",
  "@type": "NursingHome",
  "@id": "https://www.sonnenhof-bern.ch/#organization",
  "name": "Sonnenhof Bern – Alters- und Pflegezentrum",
  "description": "Private senior residence in Bern offering 120 residential places, 24/7 nursing care, specialized dementia units and day-care services since 1998.",
  "address": { "@type": "PostalAddress", "streetAddress": "Musterstrasse 12",
    "addressLocality": "Bern", "postalCode": "3007", "addressCountry": "CH" },
  "geo": { "@type": "GeoCoordinates", "latitude": 46.9480, "longitude": 7.4474 },
  "telephone": "+41 31 000 00 00",
  "priceRange": "CHF 6'500 – CHF 12'000 / month",
}
```

```

"sameAs": [
  "https://www.wikidata.org/wiki/Q000000",
  "https://www.google.com/maps/place/...",
  "https://www.local.ch/de/d/...",
  "https://www.comparis.ch/gesundheit/arzt/...",
  "https://www.facebook.com/sonnenhofbern",
  "https://www.linkedin.com/company/sonnenhof-bern"
],
"aggregateRating": { "@type": "AggregateRating", "ratingValue": "4.8", "reviewCount": "127"
}
}

```

Plus **Service** schema per offering (residential, dementia, day care, rehabilitation), **FAQPage** on every service page, **BreadcrumbList** site-wide.

Phase 2 — Become the consensus answer (Days 31–60)

5 Content cluster per language

The fan-out map: Kosten Altersheim Bern / coût EMS Berne · care levels explained · dementia guide · "Altersheim vs. Pflegeheim vs. betreutes Wohnen" · family relocation checklist · neighborhood pages (Bümpliz, Kirchenfeld, Breitenrain). Target 30–60 pages/language per quarter, each with unique data.

6 Proprietary data PR

"Berner Pflegekosten-Report 2026" — original cost/availability data → Swiss wire (SDA/Keystone) + outreach to Berner Zeitung, Bund, 20 Minuten, health trade press. Wire + syndication = the +239% citation-lift mechanic.

7 Review engineering

Systematic Google + Facebook review flow from families (Facebook reviews feed Bing Places → ChatGPT); respond to 100%. Target: 100+ Google reviews at 4.7+, steady 5–10/month velocity, keyword-rich text ("dementia", "loving staff") — review *text* is what AI quotes.

8 Listicle & directory seeding

Presence on every "best nursing homes Bern" page · fully completed comparis profile (its ranking factors: online booking, profile photo, accepting new patients) · local.ch/search.ch premium profiles.

Phase 3 — Multilingual dominance & proof (Days 61–90)

9 Localization

FR (optionally IT) full localization with hreflang; FR-language PR angle for Romandie media.

10 Claude-targeted asset

A genuinely deep, dated, citation-rich guide — *"The Complete Guide to Senior Care in Bern (2026): Types, Costs, KVG/EL Financing, and How to Choose"* — exactly the /blog/ guide format Claude cites.

11 Monitor & iterate

Weekly trilingual prompt monitoring (40-prompt library: 15 DE / 15 FR / 10 EN); iterate on whichever engine lags; refresh pages every 30 days.

12 Report & convert

SOV per engine per language · AI-referral sessions · tour bookings attributed to AI answers.

REALISTIC TIMELINE

First citations from month ~3 (matches HeyLead's own published expectation and the 4–8 week lag for retrieval-based engines); Gemini/AI Overviews depend on organic ranking maturing — 3–6 months in lower-competition Swiss local markets.

4.3 The Swiss / EU Source Ecosystem You Must Cover

Table 11 — Swiss citation ecosystem checklist

Category	Properties
Core directories	local.ch · search.ch (the two dominant Swiss directories — Bing and Google both ingest them) · help.ch
Healthcare-specific	comparis.ch medical directory (40,000+ providers, ranked by objective criteria) · doktor.ch · medicosearch · oneDoc · plus the FOPH (BAG) registry data aggregators pull from
Reviews	Google (primary) · Facebook (feeds Bing Places) · Trustpilot where applicable
Media / PR	Berner Zeitung · Der Bund · 20 Minuten · NZZ · Handelszeitung · Swiss wire distribution
Community	Reddit r/bern, r/Switzerland (expat families ask eldercare questions in English) · English Forum Switzerland — high-value for the EN query family
Registries	Zefix (Swiss commercial registry — free entity corroboration) · Crunchbase · LinkedIn

A Appendix A — Key Benchmarks & Data Points

Table 12 — Benchmark reference card (all third-party studies; treat as directional)

Finding	Value	Source context
Max visibility lift from GEO methods	+40%	Princeton / Georgia Tech / IIT Delhi, KDD 2024
Relative lift from adding source citations (mid-ranked pages)	+115%	Same study
Retrieved pages actually cited by ChatGPT	~15%	AirOps dataset
ChatGPT prompts triggering web search — overall / local	31% / 59%	AirOps
AIO citations from top-10 organic pages	93.67%	Multiple citation studies
Citation multiplier: content updated <30 days	3.2×	ZipTie citation analysis
FAQ-schema pages citation likelihood	2.8×	Same
Top-3 citation rate: schema vs no schema	47% vs 28%	Same

Correlation with AI visibility: brand mentions / backlinks	0.664 / 0.218	Ahrefs, 75k brands
AI citations from earned (non-paid) sources	95%	Muck Rack, 1M+ citations
Median citation lift from article syndication	+239%	Stacker × Scrunch, 2026
Share of LLM citations from the first 30% of a page	44.2%	Citation-position studies
AI-referral conversion: ChatGPT / Claude / Perplexity / Google organic	~15% / 16.8% / 10.5% / 1.76%	Referral analytics studies
B2B sites blocking ≥1 major AI bot	41%	Crawler audits 2025–26
Marketers tracking AI search performance	14%	First-mover window still open

B Appendix B — Tooling Stack: Build vs. Buy

Table 13 — Build vs. buy reference

Function	Buy	Build / free alternative
Multi-engine monitoring	Profound (\$2k+/mo ent.) · Peec (€89+) · Otterly (\$29+)	API workers + Postgres + Looker Studio (Section 3.3)
Prompt research	Profound Prompt Volumes · Otterly Prompt Research	AlsoAsked · AnswerThePublic · manual engine probing
Crawlability / tech audit	Otterly Crawlability Checker · Scrunch	Manual robots.txt audit · Screaming Frog · Rich Results Test
AI bot analytics	Profound Agent Analytics · Cloudflare AI metrics	Server-log grep for bot user agents (free)
Entity building	—	Wikidata · Crunchbase · GBP · Bing Places (all free)
Citation / PR execution	Muck Rack · Qwoted · Featured · wire services	Manual outreach + HARO-successors
AI referral tracking	—	GA4 custom channel with AI-referrer regex (free)

METHODOLOGY NOTE

All weightings, percentages and rankings in this report derive from published third-party citation studies and reverse-engineering research (AirOps, Ahrefs, Muck Rack, Whitespark, SE Ranking, Princeton GEO-bench, and practitioner analyses). No AI vendor publishes its ranking formula; treat every number as directional and re-validate quarterly — citation ecosystems shift fast (cf. Reddit's 60% → 10% citation swing in two weeks).

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